

Reg.No : 192525047

Name : G.Karthik Reddy

CSA0609 - DESIGN AND ANALYSIS OF ALGORITHM

Champagne Tower

1. Aim

To find the amount of champagne present in a particular glass of the tower using **Dynamic Programming**.

2. Algorithm

1. Pour champagne into the top glass.
2. If a glass contains more than 1 cup, the extra champagne flows equally to the two glasses below.
3. Continue this process row by row.
4. Return the amount in the required glass.

3. Pseudocode

Start

Put poured champagne in top glass

For each row:

For each glass:

If amount > 1:

Find extra champagne

Divide extra into two parts

Add to glasses below

Return amount in required glass

Stop

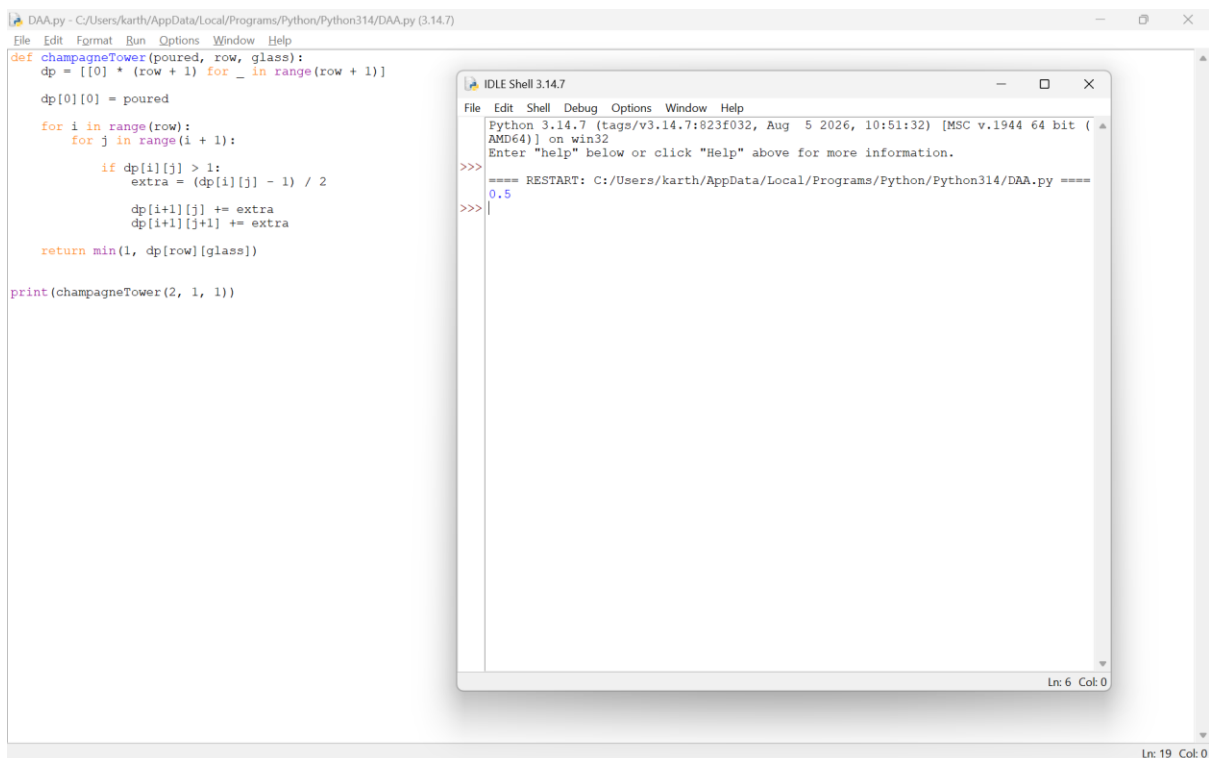
4. Simple Python Code

```
def champagneTower(poured, row, glass):  
    dp = [[0] * (row + 1) for _ in range(row + 1)]  
    dp[0][0] = poured  
    for i in range(row):  
        for j in range(i + 1):  
            if dp[i][j] > 1:  
                extra = (dp[i][j] - 1) / 2  
                dp[i+1][j] += extra  
                dp[i+1][j+1] += extra  
    return min(1, dp[row][glass])  
print(champagneTower(2, 1, 1))
```

5. Output

0.5

6. Output Screenshot



7. Complexity Analysis

- **Time Complexity:** $O(\text{row}^2)$
- **Space Complexity:** $O(\text{row}^2)$

8. Result

The amount of champagne in the specified glass was successfully calculated using **Dynamic Programming**.